Theory of Semiconductor

There are two kind of semiconductors.

(i) Pure semiconductors are called intrinsic semiconductors.

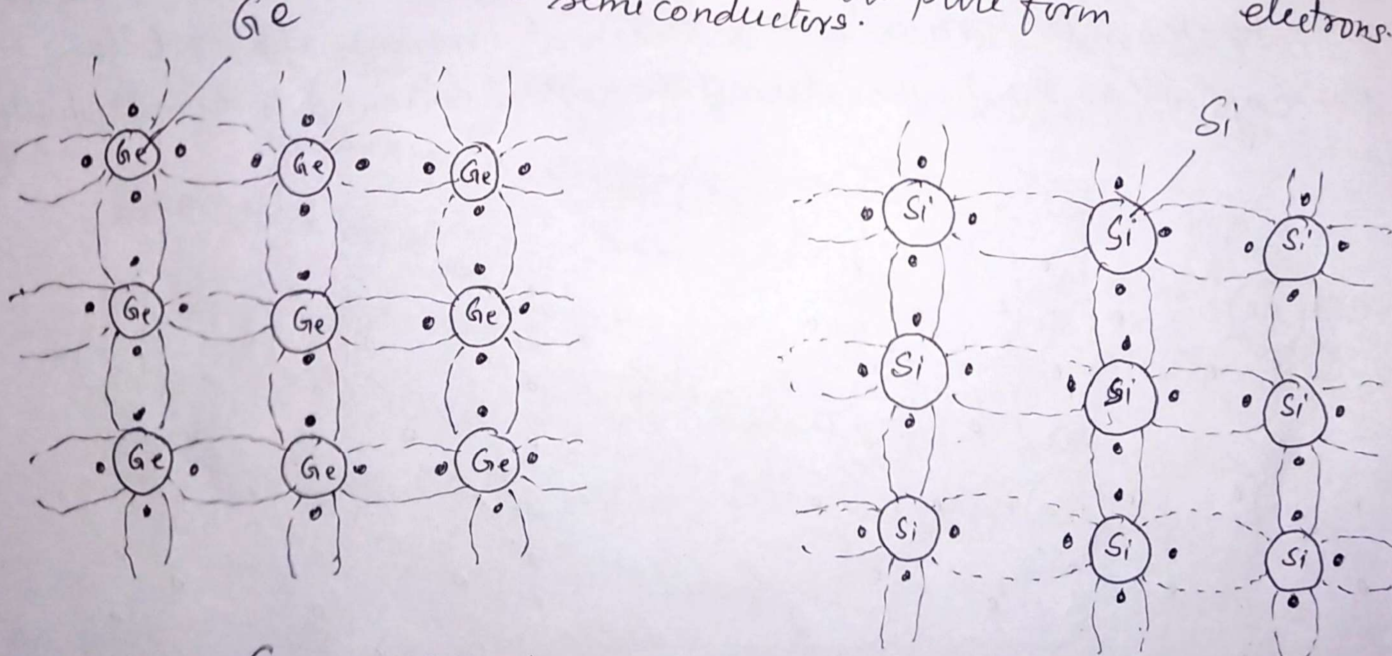
(ii) Impure semiconductors are called extrinsic semiconductors.

Intrinsic Semiconductors (Ex Si & Ge)

Si (Silicon) 14 - $1s^2 2s^2 2p^6 \underline{3s^2 3p^2} \rightarrow$ 4 Valence electrons

Ge (Germanium) 32 - $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} \underline{4s^2 4p^2} \rightarrow$ 4 Valence electrons.

The semiconductors like Ge & Si in its pure form are called intrinsic semiconductors.



Germanium

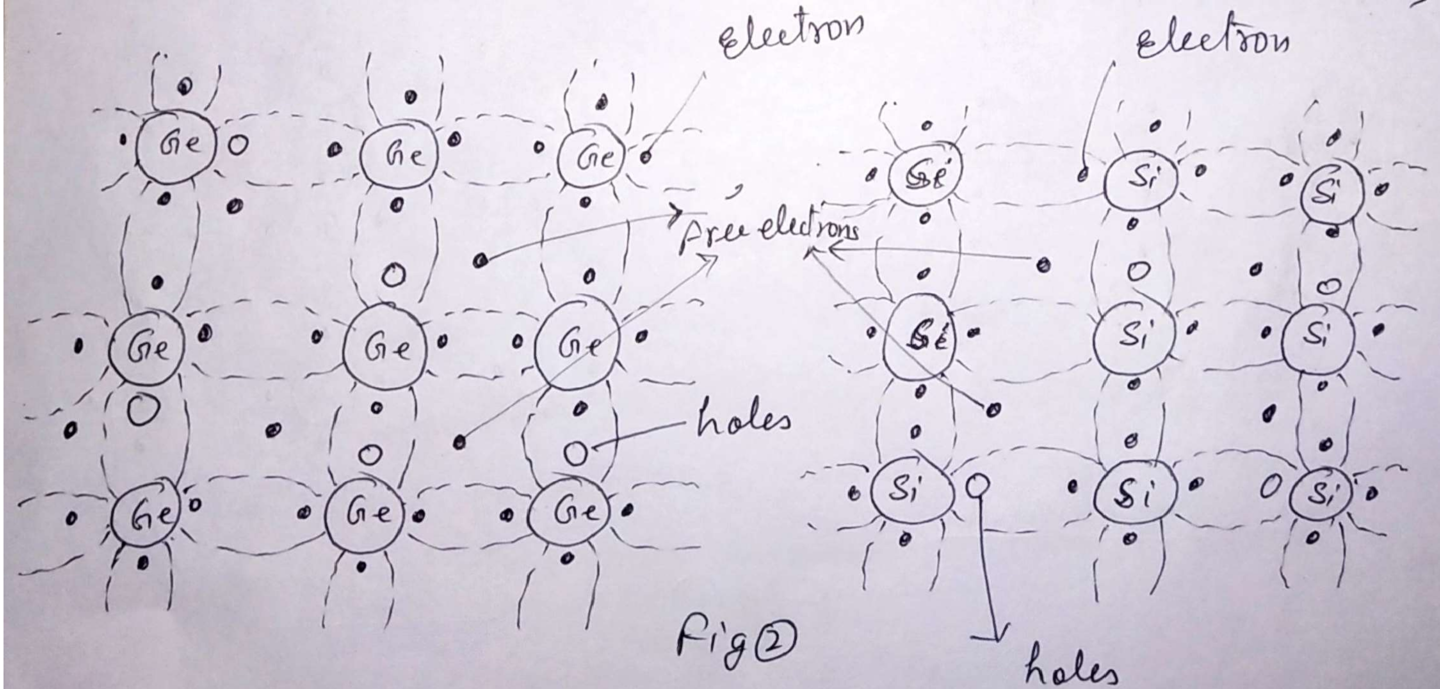
Silicon

Fig 1

Germanium and Silicon are two most important semiconductors used in electronic devices. The Germanium (Ge) and Silicon (Si) has four valence

electrons. Each Ge & Si atoms form a covalent bond with its neighbouring atoms. The symbolic structure of Ge & Si is shown in the fig①.
Since all valence electrons are bound with neighbouring ~~...~~ atoms and no ~~electron~~ free electrons are available, at ~~... at low temperatures. As soon as the temperature~~ (0°K) ~~increases and reached at room temperature the ideal structure~~
low temperature (0°K) ⇒
⇒ and the Ge & Si crystals behaves like insulators.

However at room temperature, some of the covalent bonds gets broken because of the thermal energy applied to the crystal (as the temperature of the crystal enhanced to room temperature)



fig②

~~some of the covalent bonds brokenup~~

and conduction is made possible as the electrons (very few in numbers) free to move in the crystal.

The energ E_g required to break the covalent bonds in Ge is $E_{(Ge)g} = 0.72\,eV$ and $E_{(Si)g} = 1.1\,eV$ for Si.

The abscence of electrons in the covalent bonds is shown in the fig 2 by a smal circle and such an incomplete covalent bond is called hole

Mechanism by which hole contribute to the conductivity

⇒ When an electron exit from the covalent bond a hole (deficiency of electron) ~~exists~~ exists. It is relatively easy for a valence electron in the neighbouring atom to leave its covalent bond and to fill the hole. An electron moving from a bond to fill a hole leaves a hole in its initial position. Hence the hole effectively moves in the direction opposit to that of electrons. This hole in its new position may now be filled by an electron from another covalent bond

⇒ In pure or intrinsic semi conductors the number of holes is equal to number of electrons.

$$n = p = n_i \quad \text{—} \; (1)$$

where n = number density of electrons, p = number density of holes

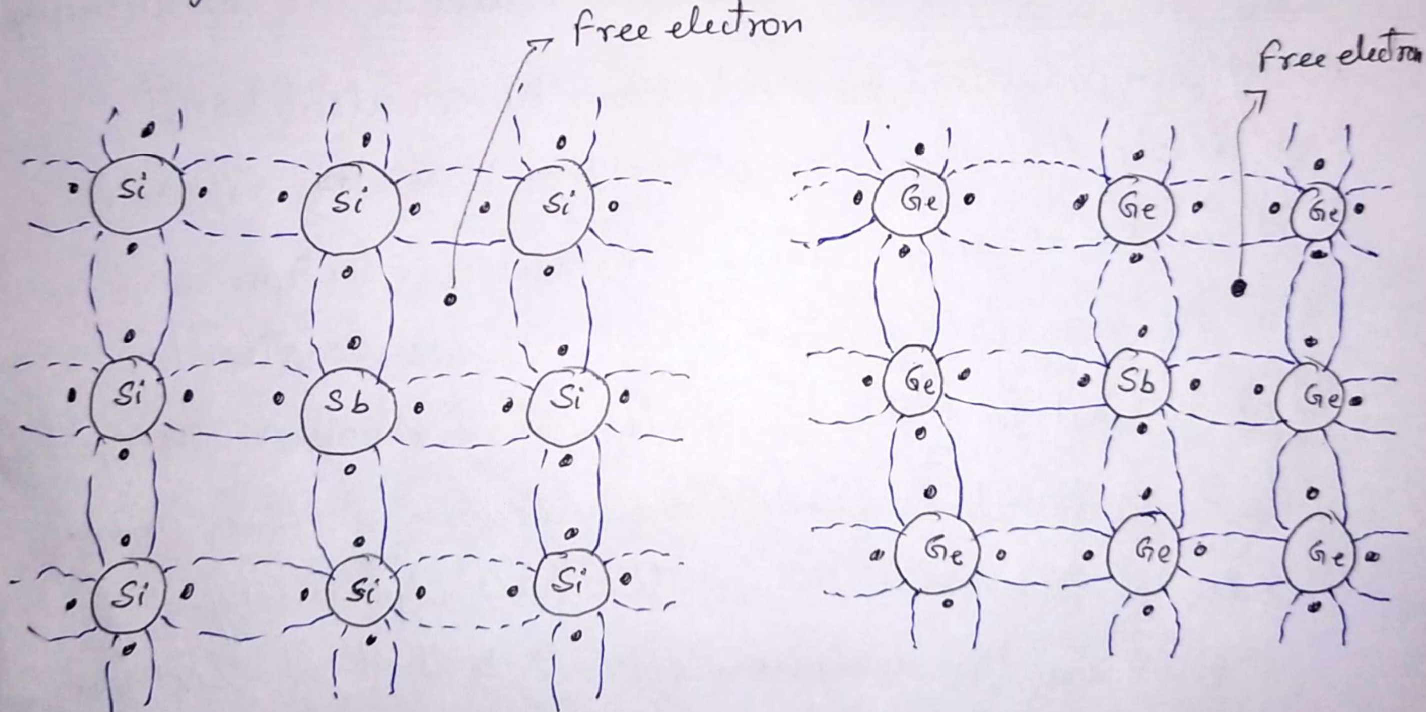
Doner & Acceptor Impurities :-

Doners :→ If a small percentage of trivalent or pentavalent atoms is added to the Ge, or Si (intrinsic semiconductor), a doped or extrinsic, semi-conductor is formed.

If the dopant has five valance electrons (Sb)

Antimony (Sb) {A = 51} $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^3$

The crystal structure will become as shown in the figure



The four of the five ~~electrons~~ valence electrons will ~~Occup~~ occupie covalent ~~bonds~~ bonds and fifth will be available for the carrier current. The energy required to detach the fifth electron from the atom is of the order of 0.01 eV for Ge & 0.05 eV for Si

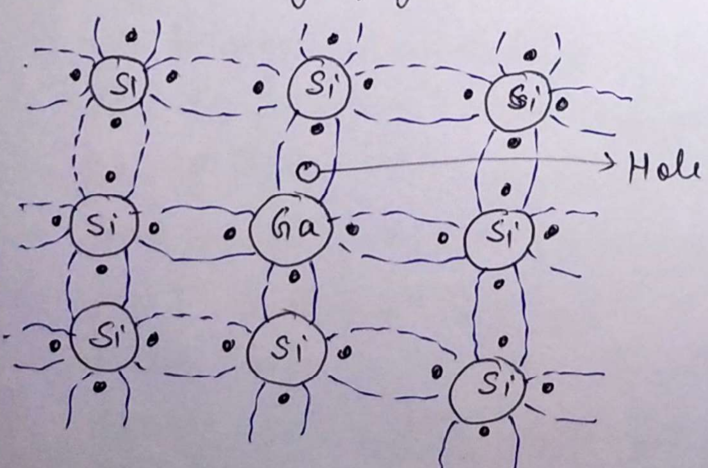
Suitable pentavalent impurities are, Atimony (Sb) Arsenic (As) & Phosphorus (P)

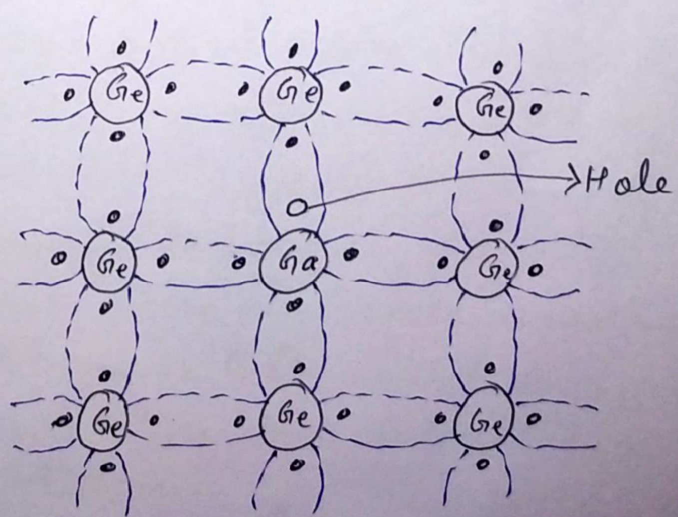
If the impurities ~~a~~ added to the intrinsic semiconductor (Ge or Si) is ~~h~~aving ~~p~~ five valency (pentavalent like Sb, As. etc) then the fifth electron of the dopant left over as free. Such type of impurities which provides the ~~et~~ extra electrons for the conduction is called doners.

~~Acceptors :→ If the trivalent impurity like~~

~~Bron~~ Boron ~~Galium (Ga), Indium In.~~

Acceptors :→ If the trivalent impurity like ~~Boron~~ Boron (B) Galium (Ga) or Indium (In) is added to the intrinsic semiconductor, only three of the covalent bonds can be filled, and the vacancy that exists in the forth bonds constitutes a hole. This situation is illustrated in the following figure.





such impurities make available positive carriers named Holes, which accept electrons. These impurities are consequently known as acceptors.

<u>n-type Semi conductor</u> :→ If doner impurity is ~~having~~ pentavalent (+5), and added to the pure Ge or Si, negative charge carriers becomes more than positive charge carrier. These type of semiconductor is called n-type semiconductor.

<u>p-type semiconductor</u> :→ If acceptor impurity is trivalent, and is added to the pure Ge or Si, then the positive charge carriers (Holes) are more than negative charge carriers, then these types of semi-conductors are called p-type semiconductors.

Fermi level in an intrinsic semiconductor

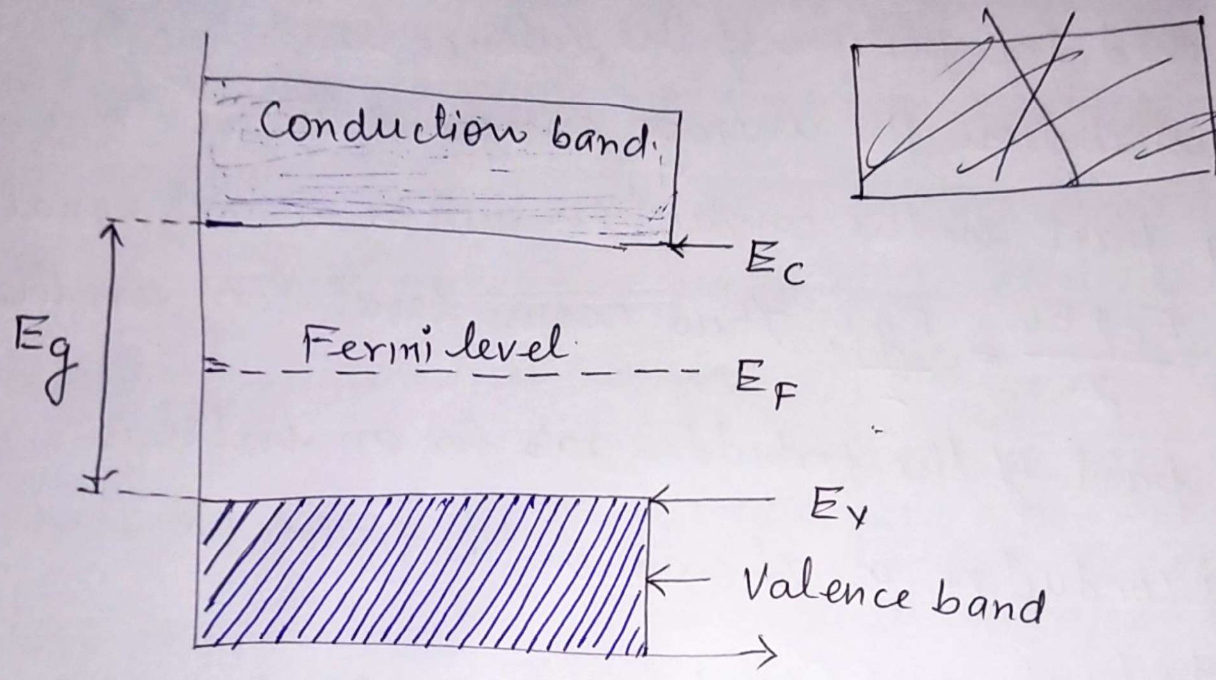


Fig (1)

Energy gap E_g is the difference between the bottom of the conduction band (E_c) and top of the valence band (E_v) as shown in the above figure. $\boxed{E_g = E_c - E_v}$

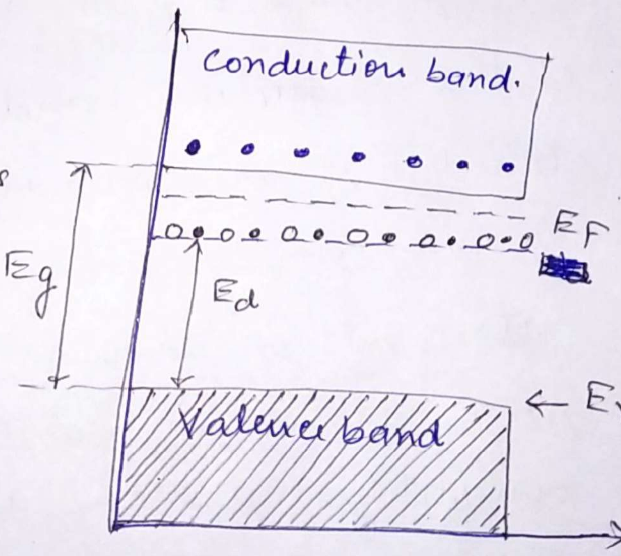
conventionally energy at the top of the valence band is taken as zero ($E_v = 0$) for refercence.

At $T = 0\,K$, energy levels in the valence band are completely filled and all energy levels in the conduction band are completely empty. But at room temperature due to thermal excitation some of the electrons at the top of the valence band are able to jump the energy gap and occupie some energy levels at the bottom part of the conduction band. These electrons returns soon to the valence band. The electrons in these set of energy levels continue to undergo excitation and deexcitation

Therefore the ~~d~~ conduction electrons between the energy levels in the bottom part of the conduction band and top portion of the valence band. Due to this distribution, the average energy of the electrons taking part in the conduction will be almost equal to $\frac{E_C + E_V}{2} = \frac{E_g}{2}$. Thus Fermi level lies in the mid part of the forbidden gap for an intrisic semiconductors as shown in the figure (1)

Impurity levels and Fermi levels in an extrinsic semiconductors

(i) n-type semiconductor :→



~~Fig(2) Fermi level~~

Fig (2) Fermi level in n-type semiconductor.

In n-type semiconductor, electrons of doner atoms are free to move and hence the posses more energy. As a result the energy level for the electrons gets elevated to a position much higher than the valence band and lie very close to the conduction band as shown in the figure (2).

The energy difference between top of the valence band and the doner level is denoted by E_d. The electrons present in the doner level are responsible for the conduction in n-type semiconductors.

Since electrons present in the doner levels stays very close to the bottom of the conduction band, therefor these electrons gets distributed between energy levels in the bottom part of the conduction band and the doner levels. The difference between these two levels, ~~as shown in figure 2~~ $(E_c - E_d)$ or $(E_g - E_d)$

$\Rightarrow$

Therefore the average energy of the electrons participating in conduction is $\frac{E_c + E_d}{2}$ or $\frac{(E_g - E_d)}{2}$

Hence the Fermi level of n-type semiconductor at low temperatures will be located in the forbidden band at the level $\frac{(E_c + E_d)}{2}$ or $\frac{(E_g - E_d)}{2}$ just below the bottom of the conduction band.

(ii) p-type semiconductors :→

In p-type semiconductors acceptor atoms (trivalent impurity) gives rise to holes which are free to move in the conduction band. Sine the energy of the holes increases opposit to the electrons and hence they occupie energy levels in the band gap & close to the valence band as shown in figure ③.

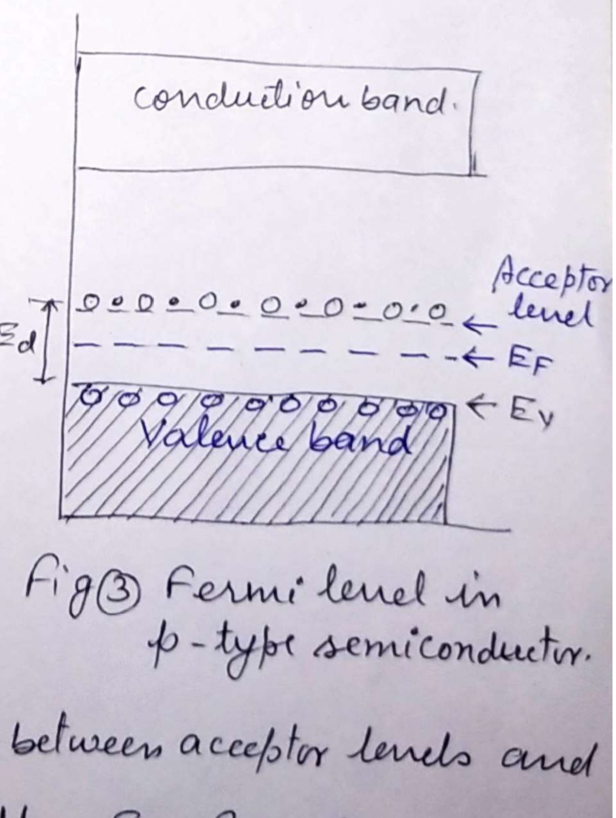


Fig ③ Fermi level in p-type semiconductor.

The energy difference between acceptor levels and top of the valence band is denoted by $E_a - E_v = E_a$ as $E_v = 0$

At low temperatures the holes participate in the conduction undergo excitation and de-excitation between acceptor levels and energy levels in the top portion of valence band.

As a result the average energy of the holes participating in conduction becomes $\frac{E_a + E_v}{2}$ or $E_a/2$

Hence the Fermi level in p-type semiconductors at low temperatures will be located in the forbidden gap at the level $\frac{E_a + E_v}{2}$ or $\frac{E_a}{2}$ from the top of the valence band.

Carrier concentration in an intrinsic semiconductor

The number of electrons in the conduction band/unit volume of the material is called the electron concentration or electron density. Similarly the number of holes in the valence band/unit volume of the material is called the hole concentration or hole density.

The number of electrons per unit volume in the energy range E & $E+dE$

$$dn_e = g(E)\,dE\,f(E) \quad -①$$

Where $g(E)\,dE$ is the number of allowed energy states per unit volume in the energy range E & $E+dE$ and $f(E)$ is fermi distribution function, represents the probability of occupation of states with energy E.

The concentration of electrons in the conduction band (n_e) can be obtained by integrating eq ① from ~~E_c to~~ $E=E_c$ to $E=\infty$

$$n_e = \int_{E_c}^{\infty} g(E)\,dE \times f(E) \quad -②$$

$$n_e = \frac{\pi}{2}\left(\frac{8m_e}{h^2}\right)^{3/2}\int_{E_c}^{\infty}\frac{E^{1/2}}{e^{\frac{E-E_F}{k_BT}}+1}\,dE \quad -③$$

In the ~~p~~ semiconductors (intrinsic) the electrons are not free but it is moving in the periodic potential so m_e should be replaced by m_e^*, the effective mass of the electron.

Further in the semiconductors, the energy of the electrons starts from E_c, hence the energy of the electron E is replaced by $(E - E_c)$. Therefore equation 3 can be written as

$$n_e = \frac{\pi}{2}\left(\frac{8m_e^*}{h^2}\right)^{3/2}\int_{E_c}^{\infty}\frac{(E-E_c)^{1/2}\,dE}{\left(e^{\frac{E-E_F}{K_BT}}+1\right)} \quad -④$$

since the Fermi ~~band~~ energy is within the forbidden energy band gap. For electrons in the conduction band $E > E_c$; If $(E_c - E_F) >> K_BT$, then

$E - E_F >> K_BT$

$$\Rightarrow \frac{(E-E_F)}{K_BT} >> 1$$

$$\Rightarrow e^{\frac{(E-E_F)}{K_BT}} + 1 \approx e^{\frac{(E-E_F)}{K_BT}} \quad -⑤$$

Now substituting the value of equation 5 in 4 we have

$$n_e = \frac{\pi}{2}\left(\frac{8m_e^*}{h^2}\right)^{3/2}\int_{E_c}^{\infty}(E-E_c)^{1/2}\,e^{-\frac{(E-E_F)}{K_BT}}\,dE \quad -⑥$$

Multiplying and dividing equation (6) by $(K_BT)^{3/2}$

$$n_e = \frac{\pi}{2}\left(\frac{8m_e^*}{h^2}\right)^{3/2}(K_BT)^{3/2}\int_{E_c}^{\infty}\left(\frac{E-E_c}{K_BT}\right)^{1/2} e^{-\frac{(E-E_c)}{K_BT}}$$

$$\times\, e^{-\frac{(E_c-E_F)}{K_BT}} \frac{dE}{K_BT}$$

$$\Rightarrow n_e = \frac{\pi}{2}\left(\frac{8m_e^*}{h^2}\right)^{3/2}(K_BT)^{3/2} e^{-\frac{(E_c-E_F)}{K_BT}}$$

$$\times \int_{E_c}^{\infty}\frac{(E-E_c)^{1/2}}{(K_BT)^{1/2}} e^{-\frac{(E-E_c)}{K_BT}} \frac{dE}{K_BT} \quad -(7)$$

Substituting

$$\frac{E-E_c}{K_BT} = x$$

$$\frac{dE}{K_BT} = dx$$

E	x
E_c	0
∞	∞

$$n_e = \frac{\pi}{2}\left(\frac{8m_e^*}{h^2}\right)^{3/2}(K_BT)^{3/2}\int_0^{\infty} x^{1/2} e^{-x}\, dx \times e^{-\frac{(E_c-E_F)}{K_BT}} \quad -(8)$$

$$\int_0^{\infty} x^{1/2} e^{-x}\, dx = \frac{\sqrt{\pi}}{2}$$

$$n_e = \frac{\pi}{2}\frac{\sqrt{\pi}}{2}\left(\frac{8m_e^*}{h^2}\right)^{3/2}(K_BT)^{3/2} \times e^{-\frac{(E_c-E_F)}{K_BT}}$$

$$\Rightarrow n_e = \frac{\pi^{3/2} \, \overset{2}{\cancel{8}} \, 2^{3/2} \left(\frac{m_e^* K_B T}{h^2} \right)^{3/2}}{4} e^{-\frac{(E_c - E_f)}{K_B T}}$$

$$\Rightarrow \boxed{n_e = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} e^{-\frac{E_c - E_f}{K_B T}}} \quad - (8)$$

suppose $N_c = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}$

$$\boxed{n_e = N_c \, e^{-\frac{E_c - E_f}{K_B T}}} \quad - (9)$$

Equations (8) & (9) represents the expression for electron concentration in conduction band of intrinsic semiconductor. Where N_c is temperature dependent material constant known as effective density of states in the conduction band.

Hole concentration in intrinsic semiconductor

Number of holes per unit volume in a semiconductor in the energy range E & E+dE.

$$dn_h = f_h(E)\, g_h(E)\, dE \quad -①$$

where $g_h(E)\,dE$ is the density of states available for holes in the energy range E & E+dE in the valence band and $f_h(E)$ is the probability of finding the holes in the energy state E.

Since the probability of having hole at some energy E in the valence band as just the same as the probability of absence of an electron at the same energy E.

Probability of absence = 1 − probability of presence

Hole probability = 1 − electron probability.

$$\Rightarrow f_h(E) = 1 - \frac{1}{\left(e^{\frac{E-E_F}{K_BT}}+1\right)} = \frac{e^{\frac{E-E_F}{K_BT}}}{\left(e^{\frac{E-E_F}{K_BT}}+1\right)}$$

$$f_h(E) = \frac{1}{1+e^{-\frac{(E-E_F)}{K_BT}}}$$

Since $K_BT << 1$

$$e^{-\frac{(E-E_F)}{K_BT}} = e^{\frac{(E_F-E)}{K_BT}} >> 1$$

$$\Rightarrow e^{\frac{E_F-E}{K_BT}} + 1 \simeq e^{\frac{(E_F-E)}{K_BT}}$$

$$f_h(E) = e^{-\frac{(E_F-E)}{(K_BT)}} = e^{\frac{E-E_F}{K_BT}}$$

$$\Rightarrow dn_h = \frac{\pi}{2}\left(\frac{8m_h^*}{h^2}\right)^{3/2} (E_V-E)^{1/2}\, dE\, e^{\frac{E-E_F}{K_BT}}$$

$$n_h = \frac{\pi}{2}\left(\frac{8m_h^*}{h^2}\right)^{3/2} \int_{-\infty}^{E_V} (E_V-E)^{1/2} e^{\frac{E-E_F}{K_BT}} dE$$

$$= \frac{\pi}{2}\left(\frac{8m_h^*}{h^2}\right)^{3/2} (K_BT)^{3/2} \int_{-\infty}^{E_V} \frac{(E_V-E)^{1/2}}{(K_BT)^{1/2}} e^{-\frac{(E_V-E)}{K_BT}} e^{\left(\frac{E_V-E_F}{K_BT}\right)} \frac{dE}{K_BT}$$

$$= \frac{\pi}{2}\left(\frac{8m_h^*}{h^2}\right)^{3/2} e^{\left(\frac{E_V-E_F}{K_BT}\right)} \int_0^{\infty} x^{1/2} e^{-x} dx$$

where $x = \frac{E_V-E}{K_BT}$

$$n_h = 2\left(\frac{2m_h^* K_BT}{h^2}\right)^{3/2} e^{-\frac{(E_F-E_V)}{K_BT}} = N_V\, e^{-\frac{(E_F-E_V)}{K_BT}} \quad -(2)$$

Relation between Fermi energy and energy gap for an intrinsic semi conductor :-

or Fermi energy in intrinsic semiconductor

For an intrinsic semiconductor, the number of electrons per unit volume in conduction band is equal to the number of holes per unit volume in valence band.

$$n_e = n_h \quad \text{—} \quad (1)$$

$$\Rightarrow \cancel{2}\left(\frac{\cancel{2}\, m_e^* \cancel{K_B T}}{\cancel{h^2}}\right)^{3/2} e^{-\frac{(E_c - E_F)}{K_B T}} = \cancel{2}\left(\frac{\cancel{2}\, m_h^* \cancel{K_B T}}{\cancel{h^2}}\right)^{3/2} e^{+\frac{(E_V - E_F)}{K_B T}}$$

$$\Rightarrow \left(m_e^*\right)^{3/2} e^{-\frac{(E_c - E_F)}{K_B T}} = \left(m_h^*\right)^{3/2} e^{+\frac{(E_V - E_F)}{K_B T}}$$

$$e^{\frac{(-E_c + E_F - E_V + E_F)}{K_B T}} = \left(\frac{m_h^*}{m_e^*}\right)^{3/2}$$

$$2E_F - (E_c + E_V) = \frac{3}{2} K_B T \log_e \left(\frac{m_h^*}{m_e^*}\right)$$

$$E_F = \left(\frac{E_c + E_V}{2}\right) + \frac{3}{4} K_B T \log_e \left(\frac{m_h^*}{m_e^*}\right)$$

$$E_F = \frac{E_c + E_v}{2} + \frac{3}{4} K_B T \log\left(\frac{m_h^*}{m_e^*}\right)$$

At absolute zero i.e. $T = 0K$

$$\boxed{E_F = \frac{E_c + E_v}{2}}$$

Sine $E_v = 0$ & $E_c = E_g$

$$\boxed{E_F = E_g/2}$$

Intrinsic carrier concentration

In an intrinsic ~~carrier~~ semiconductors there will be continuous production of electron hole pairs due to thermal energy. At the same time there will be decrease in numbers of electrons and holes due to electron-hole recombi-nation. As long as temperature remains constant the electron hole concentration will also remain constant. Therefore at temperature TK $n_e = n_h = n_i$

Therefore the product of concentration

$$n_i^2 = n_e n_h \quad \text{—} \ (1)$$

$$n_i^2 = N_c N_v \, e^{-\frac{(E_c - E_F)}{K_B T}} \, e^{\frac{(E_v - E_f)}{K_B T}}$$

$$n_i^2 = N_c N_v \, e^{\frac{-(E_c - E_v)}{K_B T}}$$

$$\Rightarrow n_i^2 = N_c N_v \, e^{-\frac{E_g}{K_B T}}$$

$$\Rightarrow n_i = \sqrt{N_c N_v} \, e^{-\frac{E_g}{2K_B T}}$$

$$\Rightarrow n_i = \left[2\left(\frac{2\pi m_e^* K_B T}{h^2}\right)^{3/2} \times 2\left(\frac{\pi 2 m_h^* K_B T}{K_B T}\right)^{3/2} \right]^{1/2} e^{-\frac{E_g}{2K_B T}}$$

$$\Rightarrow n_i = 2\left(\frac{2\pi K_B T}{h^2}\right)^{3/2} \left(m_e^* m_h^*\right)^{3/4} e^{-E_g/K_B T} \quad —(1)$$

$$\Rightarrow n_i = C\, T^{3/2} \, e^{-E_g/2K_B T} \quad —(2)$$

where $C = 2\left(\frac{2\pi K_B}{h^2}\right)^{3/2} \left(m_e^* m_h^*\right)^{3/4}$

Equation (1) & (2) represents the charge carrier concentrations in an intrinsic semiconductor.